

Lecture 6: Rheology – Measuring “Softness” and Time-Dependent Response

Key Topics

What is Rheology?

Definition: The study of flow and deformation of matter

Central Question: How do we quantify the softness or flow behavior of a material?

Focus: - Experimental rheology - Storage and loss moduli (G' , G'') - Viscosity as function of shear rate or frequency
- Relaxation timescales

Storage and Loss Moduli

Linear Viscoelastic Characterization

Method: Small oscillatory deformations

Apply oscillating strain:

$$\gamma(t) = \gamma_0 \sin(\omega t)$$

Measure resulting stress:

$$\sigma(t) = \gamma_0 [G' \sin(\omega t) + G'' \cos(\omega t)]$$

Storage Modulus (G')

Physical Meaning: - In-phase stress response - Energy stored like a spring - Measure of material stiffness - Elastic component

When G' dominates: - Material behaves solid-like - Most energy stored and recovered - Little energy dissipation

Loss Modulus (G'')

Physical Meaning: - Out-of-phase stress response - Energy dissipated as heat like a dashpot - Viscous component
- Energy lost per cycle

When G'' dominates: - Material behaves fluid-like - Energy dissipated as heat - Flow-dominated behavior

Phase Angle and Loss Tangent

Phase Angle (δ): - Lag between stress and strain - $\delta = 0^\circ \rightarrow$ Purely elastic (all energy stored) - $\delta = 90^\circ \rightarrow$ Purely viscous (all energy lost)

Loss Tangent:

$$\tan \delta = \frac{G''}{G'}$$

Interpretation: - $\tan \delta \gg 1$: Very damped, fluid-like - $\tan \delta \ll 1$: Elastic, solid-like - $\tan \delta \sim 1$: Balanced viscoelastic behavior

Rheometry Techniques

1. Rotational (Shear) Rheometer

Common Configuration: - Material sandwiched between plates - Or cone-and-plate geometry - Apply controlled strain or stress

Measurements:

Steady Shear: - Measure viscosity: $\eta = (\text{shear stress})/(\text{strain rate})$ - Flow curves - Shear-thinning or thickening behavior

Oscillatory Shear: - Measure G' and G'' - Frequency sweeps - Amplitude sweeps

Cone-Plate Geometry: - Advantage: Uniform shear across radius - Precise measurements - Common in research

2. Creep Test

Procedure: 1. Apply constant stress 2. Watch strain vs time

Results by Material Type:

Elastic Solid: - Immediate strain - Then no further creep - Constant strain (if truly Hookean)

Viscous Fluid: - Strain increases linearly forever - No asymptote - Slope = $1/\eta$

Viscoelastic Material: - Kelvin-Voigt behavior: Immediate elastic jump + slow creep $\rightarrow$ asymptote - Maxwell behavior: Indefinite creep (has liquid component)

Information Gained: - Directly measures compliance (how “soft”) - Time-dependent response - Viscosity at long times

3. Stress Relaxation Test

Procedure: 1. Apply quick fixed strain 2. Hold constant 3. Measure stress decay over time

Results by Material Type:

Purely Elastic: - Constant stress (no decay)

Viscoelastic: - Maxwell fluid: Decays to zero eventually

$$\sigma(t) = \sigma_0 e^{-t/\tau}$$

- With permanent network: Decays to plateau

Information Gained: - Relaxation time τ - Material's internal timescale - Separates elastic (short-term) and viscous (long-term)

4. Dynamic Mechanical Analysis (DMA)

Application: Solid samples (polymer bars, films)

Method: - Oscillatory loading - Often in tension or bending - Similar concept to rotational rheometry

Advantages: - For materials that can't flow into gap - Temperature sweeps - Glass transition measurements

5. Other Methods

Capillary Viscometers: - For simple fluids - Measure flow through tube - Use Poiseuille's law - Older method but still useful

Interpreting G' and G''

Practical Meanings

$G' \gg G''$ (Storage dominates): - Material is gel-like or solid-like - Can support deforming load elastically - Barely flows - Example: Firm gel, rubber

$G'' \gg G'$ (Loss dominates): - Material is liquid-like - Will flow and not hold shape - Most energy dissipated - Example: Viscous liquid, honey

$G' \approx G''$ (Crossover): - Critical point - Marks characteristic timescale - Material transitions from solid to liquid behavior - Important for gel point determination

Frequency Dependence

Many Soft Materials: - $G' < G''$ at low frequencies (fluid-like, long times) - $G' > G''$ at high frequencies (solid-like, short times) - Crossover at $\omega \approx 1/\tau$ (relaxation time)

Physical Interpretation: - High frequency (fast deformation): No time to relax $\rightarrow$ solid - Low frequency (slow deformation): Plenty of time to relax $\rightarrow$ liquid

Example Plot: - X-axis: Frequency (ω) - Y-axis: G', G'' (log scale) - Show crossover point - Identify relaxation time from crossover

Quantifying Softness

For Solids: Modulus

Young's Modulus or Shear Modulus:

Measure of Stiffness: - High modulus $\rightarrow$ stiff $\rightarrow$ "hard" - Low modulus $\rightarrow$ compliant $\rightarrow$ "soft"

Examples: - Hard plastic: $E \sim 3$ GPa - Rubber: $E \sim 1$ MPa (1000 $\times$ softer) - Soft silicone gel: $E \sim 50$ kPa (60 $\times$ softer than rubber) - Brain tissue: $E \sim 1$ kPa (extremely soft)

Biological Relevance: - Tissue engineering cares deeply about modulus - Cells respond to substrate stiffness - Preview of Engler et al. paper (Lectures 7-8)

For Fluids: Viscosity

Measure of Flow Resistance: - Low viscosity $\rightarrow$ flows easily - High viscosity $\rightarrow$ flows slowly

Examples (at room temperature): - Water: $\eta \sim 1$ mPa $\cdot$ s (very fluid) - Blood: $\eta \sim 3$ -4 mPa $\cdot$ s - Olive oil: $\eta \sim 100$ mPa $\cdot$ s - Honey: $\eta \sim 10,000$ mPa $\cdot$ s = 10 Pa $\cdot$ s - Glycerol: $\eta \sim 1000$ mPa $\cdot$ s = 1 Pa $\cdot$ s

Non-Newtonian Behavior

Shear-Thinning: - Viscosity decreases with shear rate - Common in polymer solutions, paints - Example: Ketchup (shaking lowers viscosity)

Shear-Thickening: - Viscosity increases with shear rate - Example: Cornstarch suspension (oobleck) - Acts solid under impact - Preview of Waitukaitis & Jaeger paper

Rheometer Detection: - Flow curve: Stress vs shear rate - Slope changes reveal non-Newtonian behavior

Frequency-Dependent Softness

Key Insight: Softness can depend on measurement frequency!

Example: Silly Putty - Measured slowly (low ω): Low effective modulus (soft, flows) - Measured quickly (high ω): High effective modulus (stiff, bounces)

Practical Implication: Must specify conditions when reporting "softness"

Examples and Case Studies

Laboratory Demonstration

Rheometer Reading: - Show real data from rheometer - Oscillatory test on known material - Point out G', G'' values - Discuss crossover frequency

Tabletop Viscoelastic: - Gelatin dessert - Quick oscillation (tapping) vs slow deformation (letting sag) - Relate to G' and G''

Jello vs Silly Putty

Jello at Room Temperature: - Jiggles (G' low, near solid/liquid boundary) - Breaks if strained too far - Gel with low modulus - $G' > G''$ but both relatively small

Silly Putty: - Frequency-dependent as discussed previously - Classic example of viscoelasticity

Polymer Dynamic Mechanical Analysis

Example: Polyvinyl chloride (PVC)

Below Glass Transition ($\sim 60^\circ\text{C}$): - $G' \gg G''$ (glassy solid) - Very stiff - Brittle

Above Glass Transition: - $G'' \gg G'$ (rubbery flow) - Much softer - Can flow

Demonstrates: - Temperature plays similar role to frequency - Time-temperature superposition

Data Example: Frequency Sweep

Generic Polymer Melt or Colloidal Paste:

Low ω : - $G'' > G'$ (fluid-like) - Material has time to flow

High ω : - $G' > G''$ (solid-like) - No time to relax

Crossover at ω_0 : - Mark: $\omega_0 = 1/\tau$ - Identifies relaxation time

Shear-Thinning Example

Ketchup Viscosity: - High viscosity at low shear (doesn't flow from bottle) - Low viscosity at high shear (flows when shaken/squeezed) - Flow curve shows decreasing viscosity with shear rate

Why: Structure breaks down under shear

Discussion Questions

1. Making Measurements

Scenario: New synthesized hydrogel

Question: Want to know if it's solid-like or liquid-like. What experiment?

Answer: - Oscillatory shear test at relevant frequency - Or simple poke test: rebounds (elastic) vs slowly indents (viscous) - Look at G' vs G''

2. Interpreting Moduli

Data: Material has $G' = 1000 \text{ Pa}$, $G'' = 200 \text{ Pa}$ at 1 Hz

Question: How would it behave if deformed rhythmically at 1 Hz?

Answer: - Mostly elastic response - Will return $\sim 80\%$ of energy - Only 20% lost - Will hold shape and only slowly relax

Follow-up: What if frequency increases where G'' rises and G' drops?

3. Measuring Viscosity Without High-End Rheometer

Question: How to measure honey viscosity?

Possible Answers: - Timed flow experiment (drip and use gravity + Poiseuille's law) - Falling ball viscometer (Stoke's law) - Drop marble in honey - Measure terminal speed - Compute viscosity

Engages: Creative thinking about measurements

4. Relaxation Time Thought Experiment

Comparison: Cold honey vs warm honey

Question: How would relaxation times differ?

Answer: - Warm honey: Lower viscosity → faster relaxation (low τ) - Cold honey: Higher viscosity → slower relaxation (high τ) - At room temp: Flows quickly - At fridge temp: Almost solid-like (τ very long)

Insight: "Colder = more solid-like" because molecular motion slows

5. Rheology in Everyday Life

Ketchup Question: Why does shaking bottle make it easier to pour?

Answer: - Shear-thinning behavior - Structure breaks down - Viscosity decreases - Easier flow

Milkshake Question: Why add polymers (like carboxymethyl cellulose) to make thicker?

Answer: - Increase zero-shear viscosity - Suspension stays uniform (slower creaming) - Gives smoother "body" - Better mouthfeel

Key Concepts Summary

Rheological Properties

Property	Symbol	Units	Meaning
Storage Modulus	G'	Pa	Elastic stiffness, energy stored
Loss Modulus	G''	Pa	Viscous dissipation, energy lost
Complex Modulus	G^*	Pa	$\ G^*\ = \sqrt{G'^2 + G''^2}$
Loss Tangent	$\tan \delta$	-	G''/G' (damping ratio)
Viscosity	η	Pa·s	Flow resistance

Experimental Methods Summary

Method	Measures	Best For
Oscillatory Shear	G', G''	Viscoelastic characterization
Steady Shear	$\eta(\dot{\gamma})$	Flow curves, non-Newtonian
Creep	$J(t)$	Compliance, long-time behavior
Stress Relaxation	$G(t)$	Relaxation time, memory
DMA	E', E''	Solid samples, temperature sweeps

Practical Insights

Why Rheological Properties Matter

Material Processing: - Polymer melt needs right viscosity to mold - Cured rubber needs right modulus for application

Formulation Stability: - Mayonnaise with higher G' holds shape on sandwich - Paint with proper shear-thinning flows on brush but not off wall

Product Performance: - Tire rubber needs damping (G'') for ride comfort - But stiffness (G') for handling

Biological Systems: - Mucus rheology affects clearance - Blood viscosity affects circulation - Cell mechanics affect function

Connecting Lab Numbers to Tactile Impressions

Why peanut butter doesn't flow from jar: - High zero-shear viscosity - Often has yield stress - Requires finite stress to initiate flow

Why shampoo thins when squeezed: - Shear-thinning behavior - Polymer network breaks down temporarily

Preparation for Next Lecture

Next lectures: Student Presentations (Lectures 7-8, 2-hour session)

Prepare: - Read selected paper carefully - Understand key concepts and methods - Prepare 10-12 minute presentation - Be ready for questions from peers

Papers span: - Historical foundations (Einstein, Onsager, de Gennes) - Novel materials (auxetics, soft robots) - Phase transitions (jamming, active matter) - Applications (mechanobiology, shear thickening)

References

1. "Storage modulus (G') and loss modulus (G'') for beginners" - Rheology Lab
2. "Basics of Dynamic Mechanical Analysis (DMA)" - Anton Paar Wiki
3. Macosko, C.W. "Rheology: Principles, Measurements, and Applications" (1994)
4. Barnes, H.A. "A Handbook of Elementary Rheology" (2000)

Previous Lecture: *Lecture 5: Elastic vs Viscous vs Viscoelastic* Next Lecture: *Lectures 7-8: Student Presentations*
Course Home: *Course Overview*

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